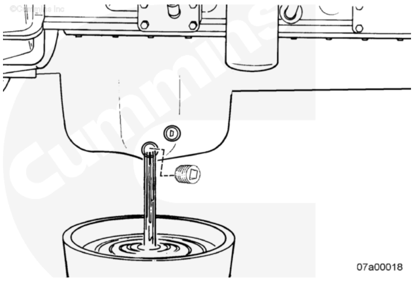
Trainings

Remove

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil.

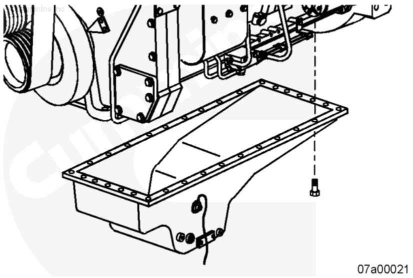


⚠ WARNING ⚠

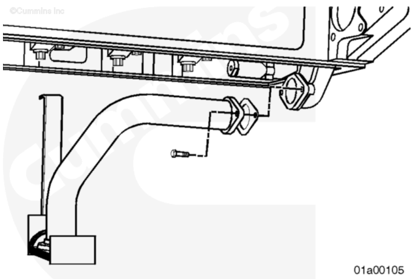
Used engine oil must be disposed of in accordance with local environmental regulations.

Drain the lubricating oil. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html)

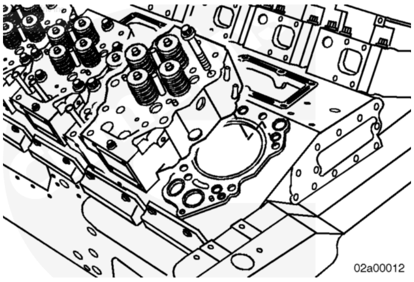
Remove the oil pan and gasket. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html)



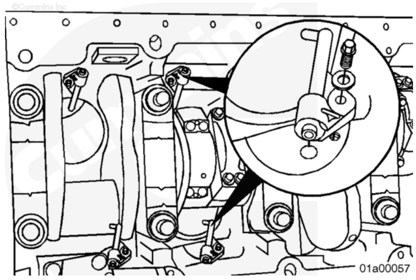
Remove the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-035-tr.html)



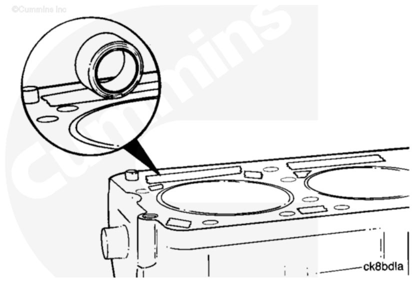
Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/57/57-002-004-tr.html)



Remove the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-046-tr.html)

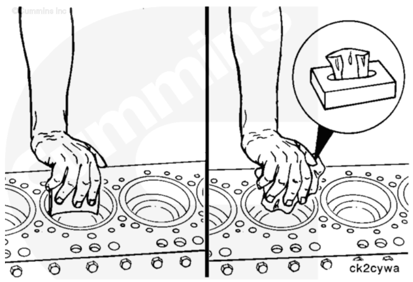


Protect the push rod galleys and coolant and oil passages from contamination.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



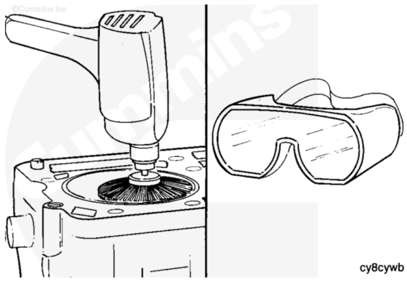
⚠ CAUTION ⚠

Do not use emery cloth or sandpaper to remove carbon from the cylinder liners. Aluminum oxide or silicon particles from emery cloth or sandpaper can cause serious engine damage. Do not use any abrasives in the ring travel area. The cylinder liner can be damaged.

Use a fine fibrous, abrasive pad such as Scotch-Brite™ 7448, Part Number 3823258, or equivalent, and solvent to remove the carbon. The carbon **must** be removed, but the surface does **not** have to appear like new metal.

⚠ WARNING ⚠

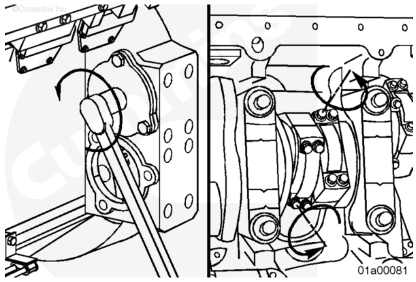
Wear eye protection to reduce the possibility of personal injury during this operation.



An alternative method to remove the carbon ridge is to use a high-quality steel wire wheel installed in a drill.

Note : Do not use a steel wire wheel of inferior quality because the wire wheel will lose steel bristles during operation and cause additional contamination.

Rotate the crankshaft to position the rod caps at bottom dead center for removal.

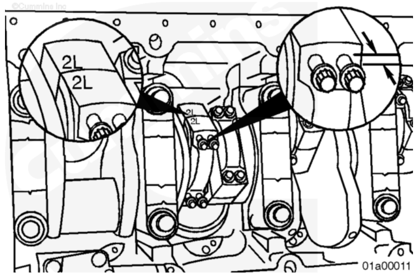


Loosen the connecting rod capscrews.

Note : Do not remove the capscrews from the rods at this time.

⚠ CAUTION ⚠

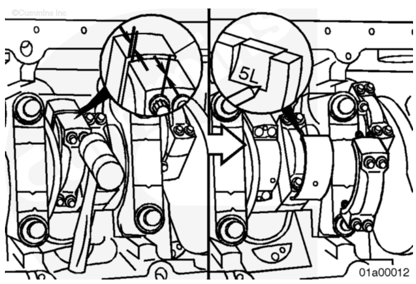
Do not use an impact wrench to loosen the connecting rod capscrews. Damage to the capscrews can occur if loosened by a power tool.



Connecting rods **must** have the cylinder number marked on both the rod and the cap on the side positioned toward the left-bank camshaft. Inspect the rods for correct markings. Use a steel stamp to mark any rod that is **not** correctly marked.

Loosen the capscrews until there is 6 mm [1/4 in] of clearance between the rod cap and the capscrew head.

Strike the connecting rod capscrews with a rubber mallet to loosen the caps from the dowel rings.



Remove the connecting rod capscrews.

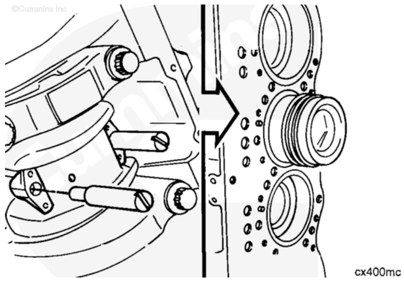
Remove the rod caps.

Remove the lower rod bearings.

Mark the cylinder number and the letter L (lower) on the flat surface of the bearing tangs.

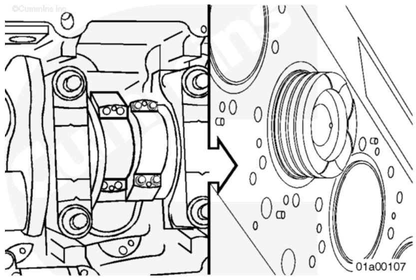
⚠ CAUTION ⚠

Be careful not to damage the inside surface of the liner with the connecting rod.



Use two 11-mm nylon connecting rod guide pins, Part Number 3163363.

Install the guide pins to prevent the rod from falling and damaging other engine components.

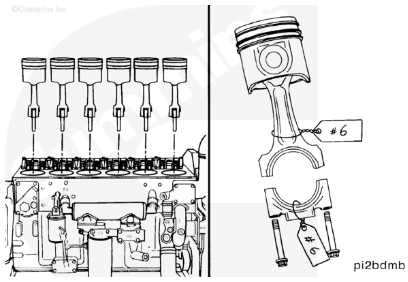


Use a T-handle piston pusher to push the rod away from the crankshaft.

Push the rod until the piston rings are outside of the top of the cylinder liner.

⚠ CAUTION ⚠

Put the piston and connecting rod assembly in a rack to reduce the possibility of damage.

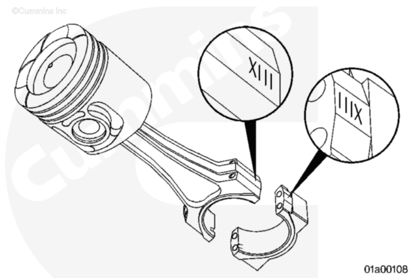


Use both hands to remove the piston and rod assembly.

The piston and connecting rod assemblies **must** be installed in the same cylinder number from which they were removed, to maintain proper fit of worn mating surfaces, if parts are used again.

Use a tag to mark each piston and rod assembly with the appropriate cylinder number.

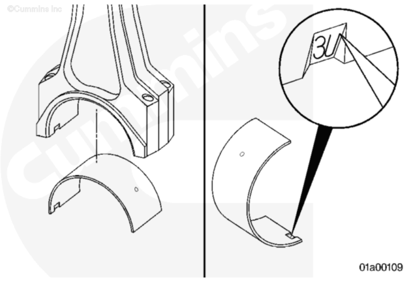
Note : Connecting rods are specific to the right or left bank.



A unique number (**not** cylinder number) is etched on the connecting rod and matching cap. When the rods and caps are installed on the engine, the numbers on the rods and caps **must** match and be installed on the same side of the engine.

Remove the upper rod bearing.

Mark the cylinder number and the letter U (upper) on the flat surface of the bearing tang.

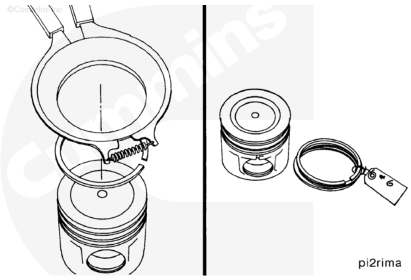


Disassemble

Use piston ring expander, Part Number 3823871, to remove the piston rings.

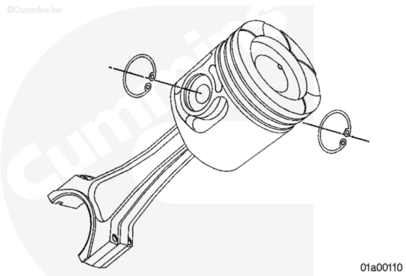
Place a tag on the rings, and record the cylinder number of the piston on the tag.

Use the following procedure for piston ring inspection. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-047.html)



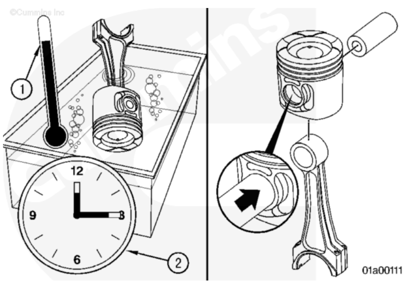
Piston and Connecting Rod - Disassembly

Use internal snap ring pliers to remove the snap rings from both sides of the piston.



⚠ CAUTION ⚠

Do not use a hammer to remove the piston pins. Force can distort the piston, causing it to seize in the liner.



Note : Mark the cylinder number from where the piston and pin were removed on the parts to make sure they are installed in the same cylinder.

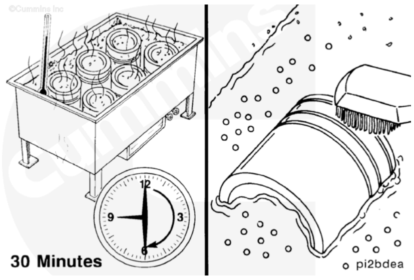
If the piston pin can **not** be easily removed by hand:

- Install the piston and rod assembly in a container of water.
- Heat the piston in boiling water for 5 minutes.

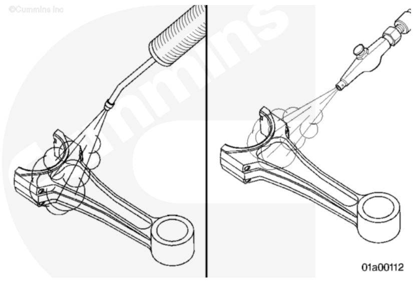
- Use a blunt tool to push the piston pin from the piston and rod assembly.

Clean

Use the following procedure for piston cleaning instructions. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-043.html)

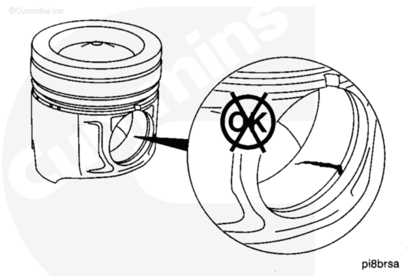


Use the following procedure for instructions on cleaning the connecting rods. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-014.html)

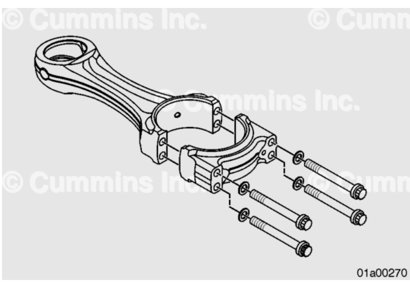


Inspect for Reuse

Use the following procedure for inspection specifications for QST30 Series engine pistons. Refer to Procedure 001-043 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-043.html)



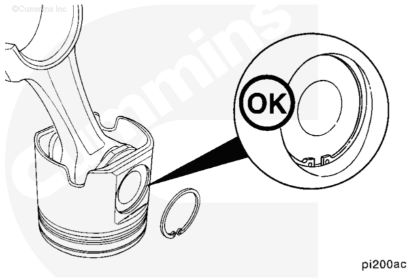
Use the following procedure for inspection specifications for QST30 Series engine connecting rods. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-014.html)



Assemble

⚠ CAUTION ⚠

The retainer snap ring must be seated completely in the piston pin groove to reduce the possibility of engine damage during engine operation.

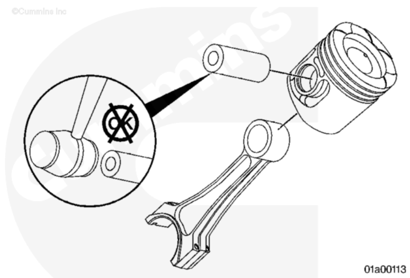


Note : The flat side of the retaining ring **must** face outward.

Install a new snap ring in one piston pin bore of each piston.

⚠ CAUTION ⚠

The left- and right-bank connecting rods in the QST30 Series engine are not interchangeable. The left-bank connecting rod is Part Number 3092935; the right-bank connecting rod is Part Number 3092932. Use of the wrong connecting rod will cause engine failure.



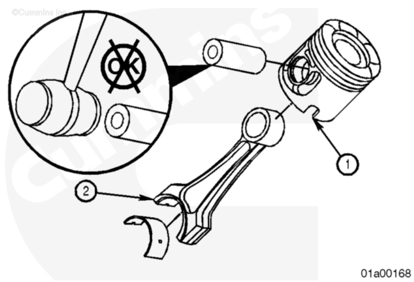
⚠ CAUTION ⚠

Do not use a hammer to install the piston pin. The piston can distort, causing it to seize in the liner.

Note : The pistons and liners are matched by size: Large (left bank) and small (right bank).

Note : If the pistons are being reused, use the matched piston and pin previously removed.

Note : It is **not** necessary to heat the pistons before assembly. The piston pin is slipfit.

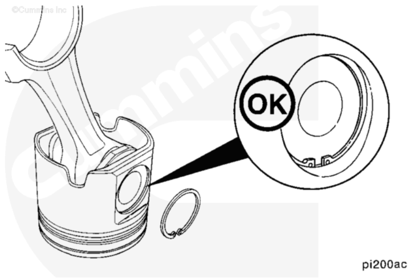


Align the piston cooling nozzle notch (1) with the beveled side of the large end of the connecting rod (2).

Lubricate the piston pin and the inner diameter of the connecting rod bushing with clean engine oil.

Install the piston pin.

Note : The flat side of the retaining ring must face outward.

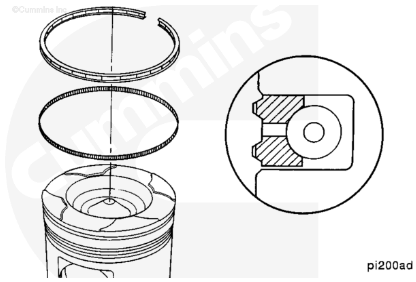


Install the second snap ring.

Rotate both snap rings to make sure they are completely installed in the ring groove.

Make sure the connecting rod moves freely.

Piston Rings - Assemble



Note : Do not install the piston rings until the pistons are assembled to the rods.

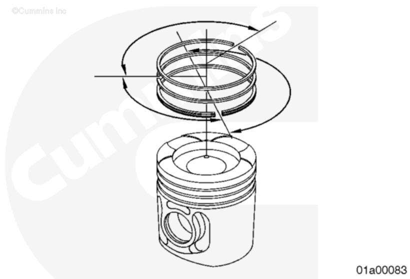
Note : The end gap of the expander ring must be 180 degrees from the end gap of the oil control ring. Do not overlap the ends of the expander ring.

Separate the oil ring expander from the oil control ring.

Fit the oil ring expander into the oil ring groove of the piston, and connect the two ends of the expander ring by inserting the guide wire into the opposite end.

Use piston ring expander, Part Number 3823871. Install the oil control ring over the expander.

Note : The stamped side of each ring must face up.



Note : Do not align the gap of a ring with the piston pin bore.

Install the second ring and the top ring on the piston, in order. Use the piston ring expander, Part Number 3823871.

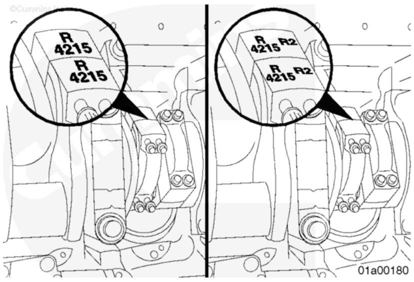
Rotate the rings until the gaps are positioned as shown (rings staggered to prevent blowby).

Install

If reusing the old connecting rods and caps, check to make sure the cylinder number is stamped on both the rod and the cap.

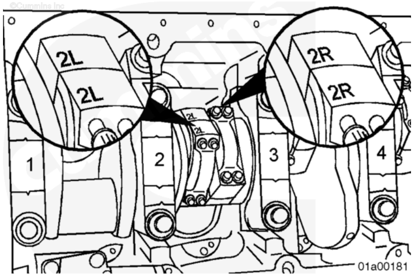
If installing new rods and caps, there will be an R or L plus a unique number (**not** a cylinder number) etched on one side of the connecting rod and mating cap.

Use a hammer and a stamp to stamp an R for the right bank or L for the left bank, plus the cylinder destination number on the same side as the etched number.



⚠ CAUTION ⚠

When installing the connecting rods, the stamped cylinder numbers on the rods (for both the right- and left-bank rods) must be positioned toward the left-bank camshaft. Engine damage can occur if the rods are installed incorrectly.



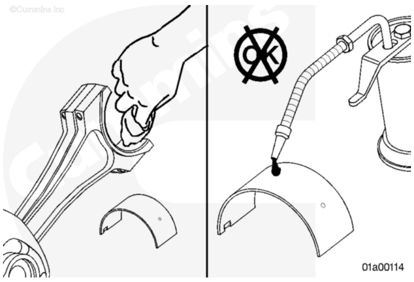
⚠ CAUTION ⚠

The cylinder block and all parts must be clean before assembly.

⚠ CAUTION ⚠

Do not lubricate the backside of the bearing shells. The operating clearance of the bearing will be reduced and the bearing can be damaged during engine operation.

Use a clean, lint-free cloth to clean the connecting rods and bearing shells.

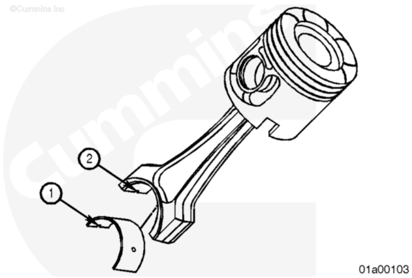


Note : The tang of the bearing shell (1) must be in the slot of the rod (2). The end of the bearing shell must be even with the cap mounting surface.

Note : Verify that all oil holes in bearing shells align with holes in the connecting rod.

If new bearings are **not** used, the used bearings **must** be installed on the same connecting rod from where they were removed.

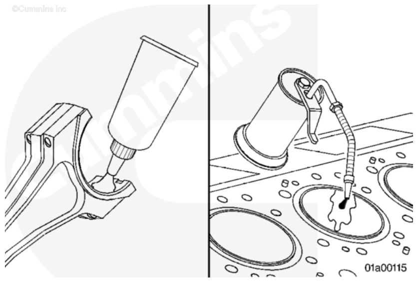
Install the upper bearing shell into the connecting rod.



Note : The entire liner bore must be lubricated.

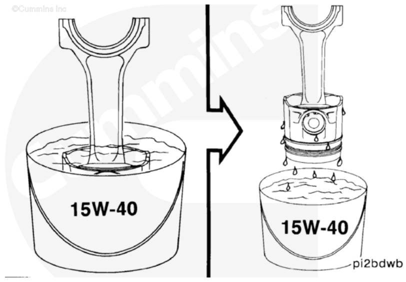
Use Lubriplate™ 105, or equivalent, to coat the inside of the bearing shell.

Apply a heavy film of clean SAE 15W-40 engine oil to the cylinder liner.



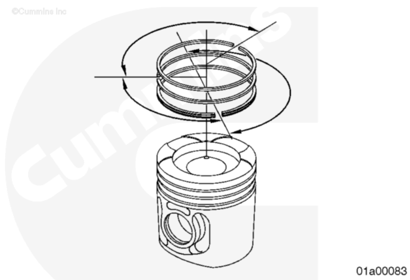
Immerse the piston and ring assembly in a container of clean SAE 15W-40 oil.

Remove the piston and ring assembly from the container and let the excess oil drain from the piston.



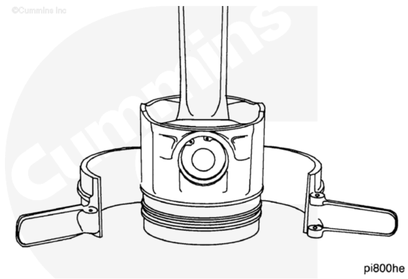
Note : The ring gap of each ring must not be aligned with the piston pin, or with any other ring. If the ring gaps are not aligned correctly, the rings will not seal properly.

Note : The end gap of the expander ring must be 180 degrees from the end gap of the oil control ring. Do not overlap the ends of the expander ring.

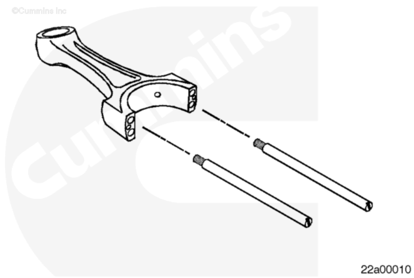


Rotate the rings to position the ring gaps as shown.

Use piston ring compressor, Part Number 4918212, to compress the rings.



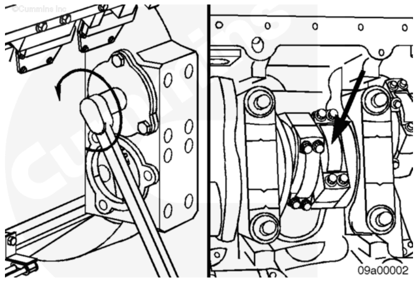
Install nylon connecting rod guide pins, Part Number 3163363.



⚠ **CAUTION** ⚠

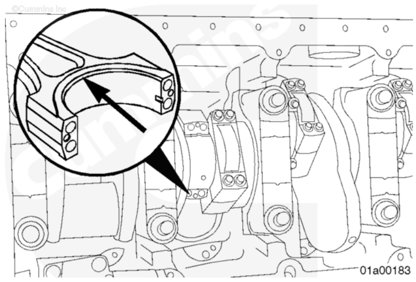
The cylinder number on the rod and cap must be the same.

Rotate the crankshaft until the connecting rod journal of the connecting rod being installed is at bottom dead center.



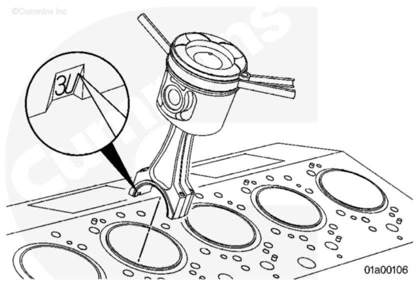
⚠ CAUTION ⚠

The connecting rod must be installed so that the chamfer on the rod is toward the fillet radius on the crankshaft rod journal. If the rod is reversed, the crankshaft can be damaged.



⚠ CAUTION ⚠

The bearing tang side of the connecting rods (for both the right- and left-bank rods) must face the left-bank camshaft or damage to the crankshaft or connecting rod can occur.



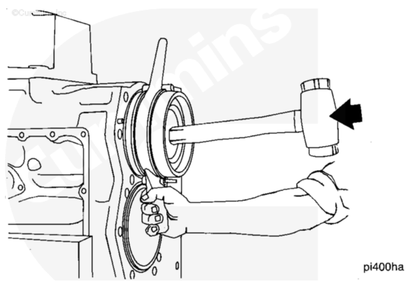
⚠ CAUTION ⚠

The piston size must match the liner size or engine damage can occur.

Insert the connecting rod into the cylinder liner with the bearing tang toward the left-bank camshaft side of the engine until the ring compressor contacts the top of the liner.

⚠ CAUTION ⚠

Do not use a metal drift to push the piston into the cylinder liner. The piston rings or cylinder liner can be damaged.

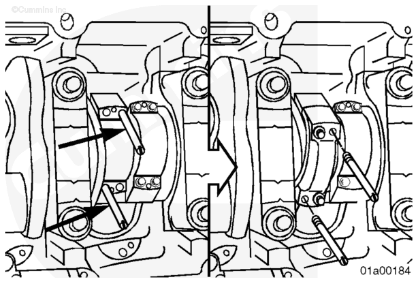


Hold the ring compressor against the cylinder liner. Push the piston through the ring compressor and into the cylinder liner. Push the piston until the top ring is completely within the cylinder liner.

Make sure the piston moves freely within the cylinder. If the piston does **not** move freely, remove the piston, and inspect for broken or damaged rings.

Use the nylon guide pins to align the connecting rod with the crankshaft while pushing the piston and rod assembly in place.

Remove the nylon guide pins.

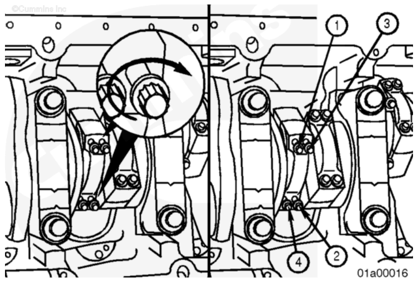


⚠ CAUTION ⚠

The connecting rod and cap must have the same number and must be installed in the proper cylinder or engine damage can occur.

⚠ CAUTION ⚠

The connecting rod cap number and rod number must be on the same side of the connecting rod to reduce the possibility of damage during engine operation.

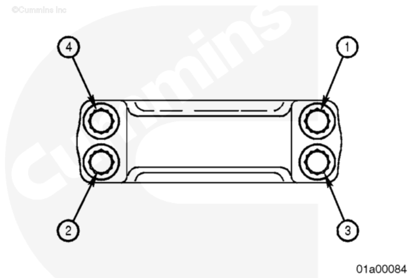


Use clean 15W-40 engine oil to lubricate the connecting rod capscrew threads and washers.

Install the connecting rod caps, capscrews, and washers.

Hand-tighten all capscrews first.

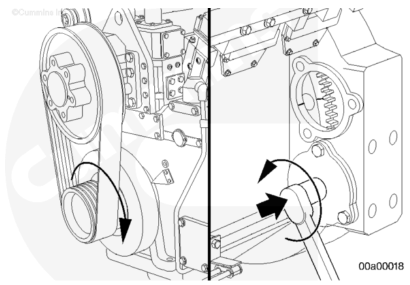
Tighten the connecting rod capscrews. Refer to Procedure 001-014 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-014.html)



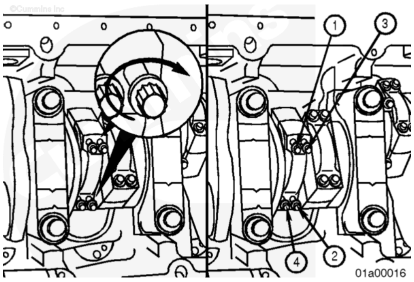
Two cylinders, right-bank and left-bank, can have the piston assembly installed and tightened without having to bar the engine (i.e., number 1R, number 1L, number 6R, and number 6L).

After completion, bar the engine over so that the number 3, number 4, number 2, and number 5 journals are at bottom dead center.

Install the pistons.



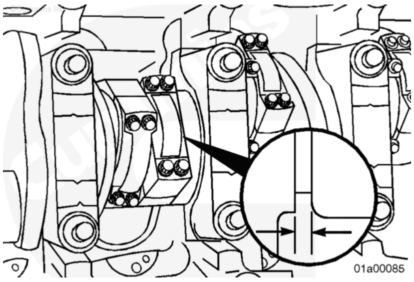
Note : It is a good service practice to go back and check all rod capscrew torques after all piston assemblies have been installed and tightened.



Note : Both rods must move freely side to side.

Measure the connecting rod side clearance.

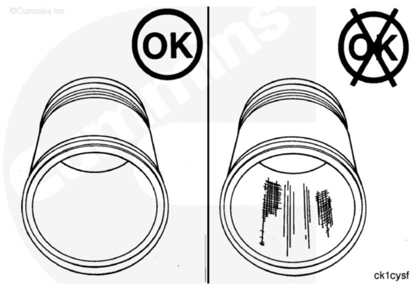
Connecting Rod Side Clearance		
mm		in
0.180	MIN	0.007
0.454	MAX	0.018



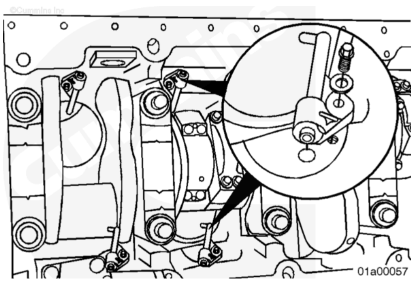
Using a barring tool, rotate the engine at least two revolutions; then inspect the cylinder liners for damage.

If damage is present, remove the piston, and inspect the rings.

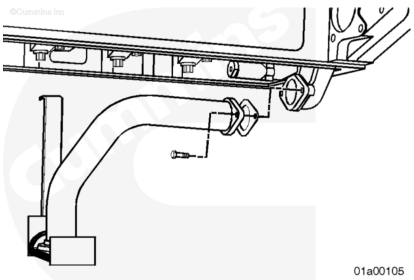
Replace if necessary.



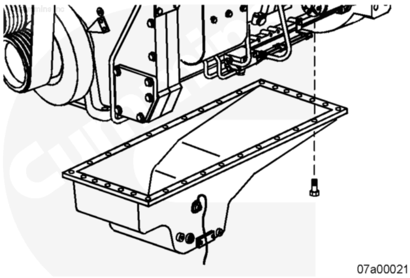
Install the piston cooling nozzles. Refer to Procedure 001-046 in Section 1. (/qs3/pubsys2/xml/en/procedures/57/57-001-046-tr.html)



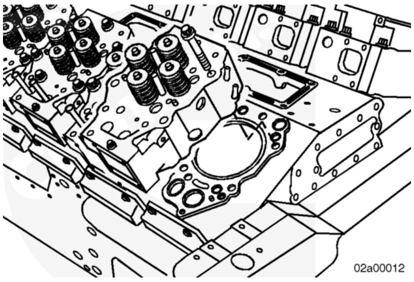
Install the oil suction tube. Refer to Procedure 007-035 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-035-tr.html)



Install the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html)

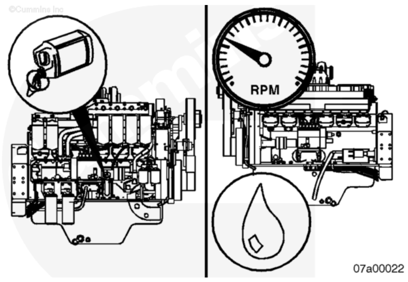


Install the cylinder heads. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/57/57-002-004-tr.html)



Fill the lubricating oil pan. Refer to Procedure 007-025 in Section 7. (/qs3/pubsys2/xml/en/procedures/57/57-007-025-tr.html)

Operate the engine to normal operating temperature and check for leaks.



Last Modified: 14-Apr-2016